Optimal Placement of Position Reconfigurable Antennas

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Slides & Codes: https://scliubit.github.io

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Outline

1. **Background**
2. **Antenna Density (Continuous)**
3. **Antenna Placement (Discrete)**
4. **Conclusions**

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2. Antenna Density (Continuous)
3. Antenna Placement (Discrete)
4. Conclusions

Background: Position Reconfigurable Antennas

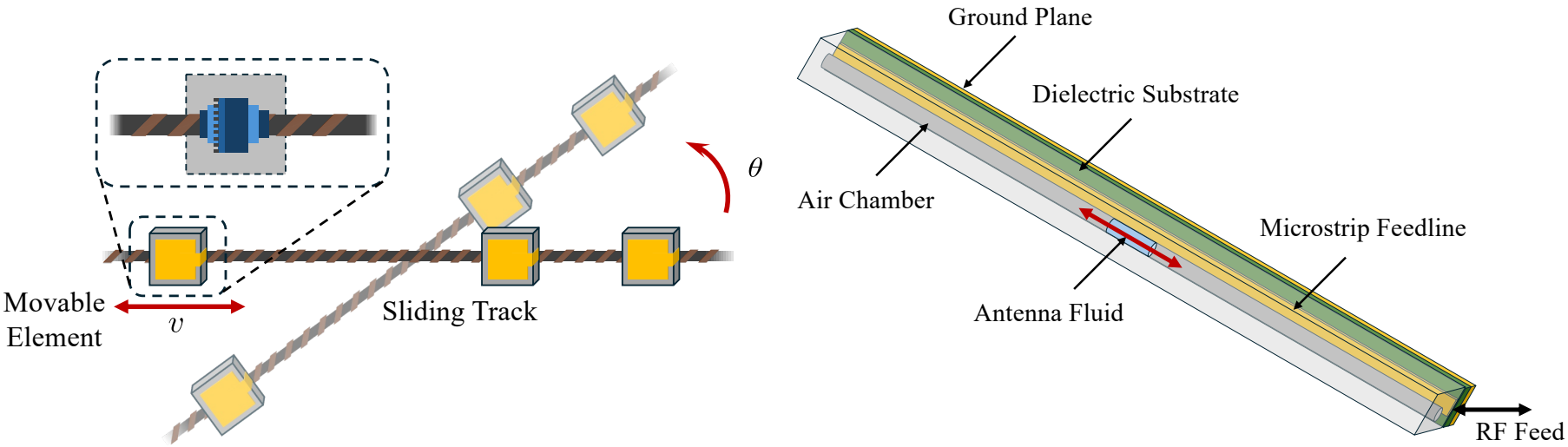


Figure 1: Movable/Pinching antennas [1] (left) and fluid antennas [2] (right).

- **Mechanically Movable**: Slow; Improve channel degree of freedom (DoF).
- **Electronically Fluid**: Fast; Obtain better channel conditions (receive power).

[1] L. Zhu, W. Ma, and R. Zhang, "Movable antennas for wireless communication: Opportunities and challenges," *IEEE Wireless Commun.*, vol. 62, no. 6, pp. 114–120, 2024.
[2] K.-K. Wong, A. Shojaeifard, K.-F. Tong, and Y. Zhang, "Fluid antenna systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1950–1962, 2021.

Literature Review

- **Movable Antenna**: Branch and Bound, SCA [3]
- **6DMA**: 3D Position and 3D Rotation, Monte Carlo [4]
- **Fluid Antenna**: Alternating Optimization [5]
- **Pinching Antenna**: MM and Penalty Dual Decomposition [6]
- ...

[3] Y. Wu, D. Xu, D. Wing Kwan Ng, W. Gerstacker, and R. Schober, "Globally optimal movable antenna-enabled multiuser communication: Discrete antenna positioning, power consumption, and imperfect csi," *IEEE Trans. Commun.*, vol. 73, no. 10, pp. 9903–9923, 2025.

[4] X. Shao, R. Zhang, Q. Jiang, and R. Schober, "6D movable antenna enhanced wireless network via discrete position and rotation optimization," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 674–687, 2025.

[5] T. Hao, C. Shi, Q. Wu, B. Xia, et al., "Fluid-antenna enhanced isac: Joint antenna positioning and dual-functional beamforming design under perfect and imperfect csi," *IEEE Trans. Veh. Technol.*, vol. 74, no. 11, pp. 17 204–17 219, 2025.

[6] X. Xu, X. Mu, Y. Liu, and A. Nallanathan, "Joint transmit and pinching beamforming for pinching antenna system (pass): Optimization-based or learning-based?" *IEEE Trans. Wireless Commun.*, vol. 25, pp. 11 449–11 464, 2026.

Motivation

- Most previous works deal with antenna placement by numerical optimization.
  - Limited physical insights
  - High computational complexity
  - Near-field Effect ignored

- In this seminar, I will introduce my work that addresses all three challenges

Representative Works

[7] S. Liu, X. Yu, J. Xu, and R. Zhang, "Near-field communication with massive movable antennas: A functional perspective," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 14 455–14 470, 2026.

[8] S. Liu, X. Yu, S. Song, and K. B. Letaief, "Near-field communication with movable antennas: An electrostatic equilibrium perspective," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 16 024–16 039, 2026.

Outline

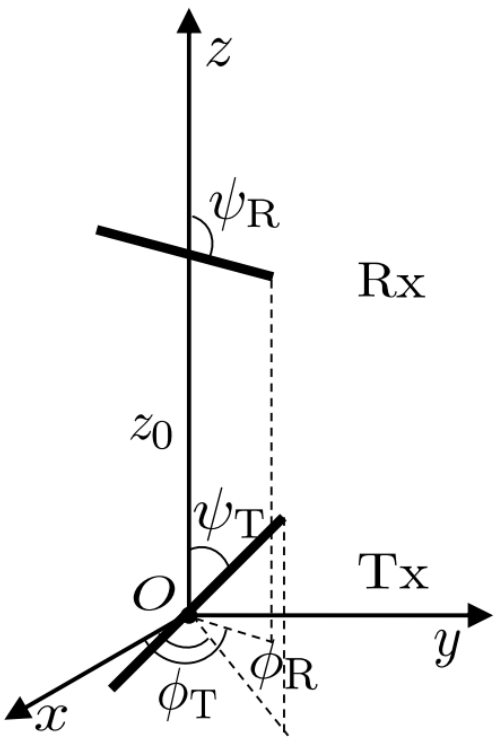
1 Background

2 **Antenna Density (Continuous)**

3 Antenna Placement (Discrete)

4 Conclusions

System Model



- Point to point wireless system, LoS-dominant channel.
- M movable elements at Tx, N fixed elements at Rx.
- Antennas move freely on a linear array with coordinate

$$\begin{cases} x_{\mathrm{T}}^{(m)} &= \frac{A_{\mathrm{T}}}{2} f(m) \sin\theta_{\mathrm{T}} \cos\phi_{\mathrm{T}} \\ y_{\mathrm{T}}^{(m)} &= \frac{A_{\mathrm{T}}}{2} f(m) \sin\theta_{\mathrm{T}} \sin\phi_{\mathrm{T}}, \quad \forall 1 \le m \le M, \\ z_{\mathrm{T}}^{(m)} &= \frac{A_{\mathrm{T}}}{2} f(m) \cos\theta_{\mathrm{T}} \end{cases}$$

where

- A_{T} is the overall aperture (move range)
- Continuous **antenna position function** (APF)

$$p = f(m)$$

denotes the normalized position $-1 \le p \le 1$ given index m.

Antenna Density Function

- The number of antenna elements within unit length

$$w(p) = \lim_{\Delta p \to 0} \frac{f^{-1}(p + \Delta p) - f^{-1}(p)}{\Delta p} = \frac{\mathrm{d}}{\mathrm{d}p} f^{-1}(p),$$

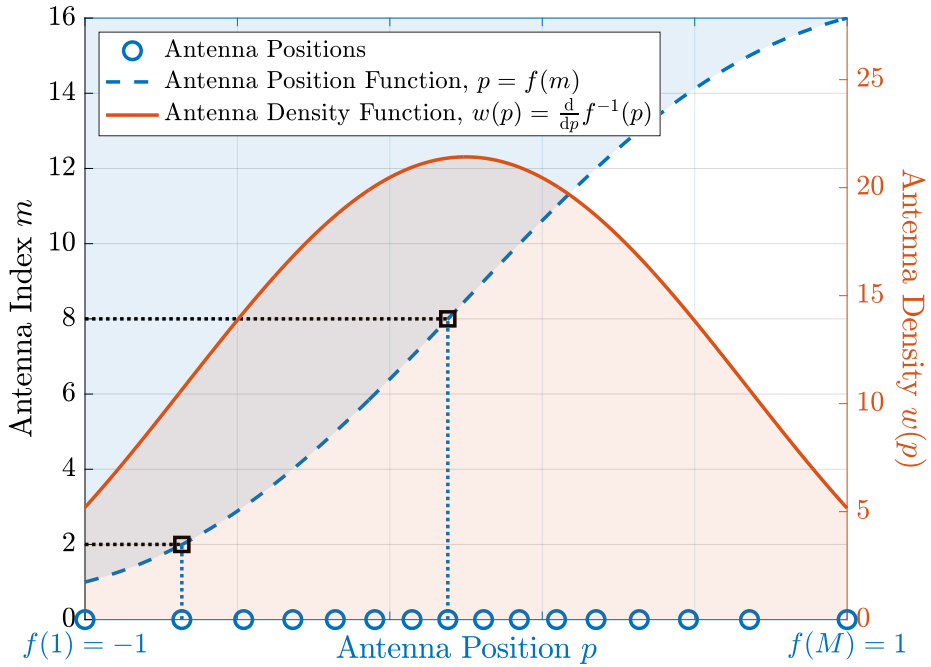


Figure 2: APF $p = f(m)$ (in blue) and ADF $w(p)$ (in orange).

Problem Formulation

- Wireless channel matrix

$$\mathbf{H}_w[m,n] = \frac{e^{-\jmath\kappa r_{m,n}}}{r_{m,n}}$$

  is a *functional* of ADF $w(p)$ (or APF $f(m)$).

- Find optimal $w(p)$ to maximize the achievable rate *functional*

$$C_w = \log_2 \det\left(\mathbf{I} + \frac{P_{\mathrm{T}}}{M\sigma_{\mathrm{n}}^2}\mathbf{H}_w\mathbf{H}_w^H\right) = \log_2 \det\left(\mathbf{T}_N\right)$$

- **Goal:** To optimize ADF $w(p)$ that maximizes **Toeplitz determinant**.

Problem Formulation

- Each Toeplitz matrix is associated with its unique generating function $s(\theta)$. It generates the column elements via Fourier transform (See [7] for more details).

- (Fresnel approx. and massive M) The generating function of $\mathbf{T}_N$ is

$$s(\theta) = 1 + \frac{2\pi\rho}{\beta z_0^2} w\left(\frac{\theta}{\beta}\right),$$

  where $\beta = \frac{\kappa A_{\mathrm{T}} A_{\mathrm{R}} \sin\theta_{\mathrm{T}} \sin\theta_{\mathrm{R}} \cos(\phi_{\mathrm{T}} - \phi_{\mathrm{R}})}{z_0(N-1)}$ is an antenna geometry-related parameter.

- **Goal transfer:** Find $s(\theta)$ that maximizes $\log_2 \det(\mathbf{T}_N)$

[7] S. Liu, X. Yu, J. Xu, and R. Zhang, "Near-field communication with massive movable antennas: A functional perspective," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 14 455–14 470, 2026.

Optimal ADF

- By leveraging the Fisher-Hartwig Theorem [9], [10], the optimal generating function (in angular domain) should be

$$s^\star(\theta) \propto |1-\cos(\theta-\beta)|^\alpha \, |1-\cos(\theta+\beta)|^\alpha, \qquad -\frac{1}{2} < \alpha < 0$$

- Transform back with proper approximation

$$w^\star(p) = \left(\frac{\gamma_\alpha}{(1-p^2)^{-2\alpha}} - \frac{\beta z_0^2}{2\pi\rho} \right) (1+\tau p)^2 \simeq \frac{\gamma_\alpha}{(1-p^2)^{-2\alpha}}$$

  where γ_α is a normalization factor, τ relates to the angle of arrival.

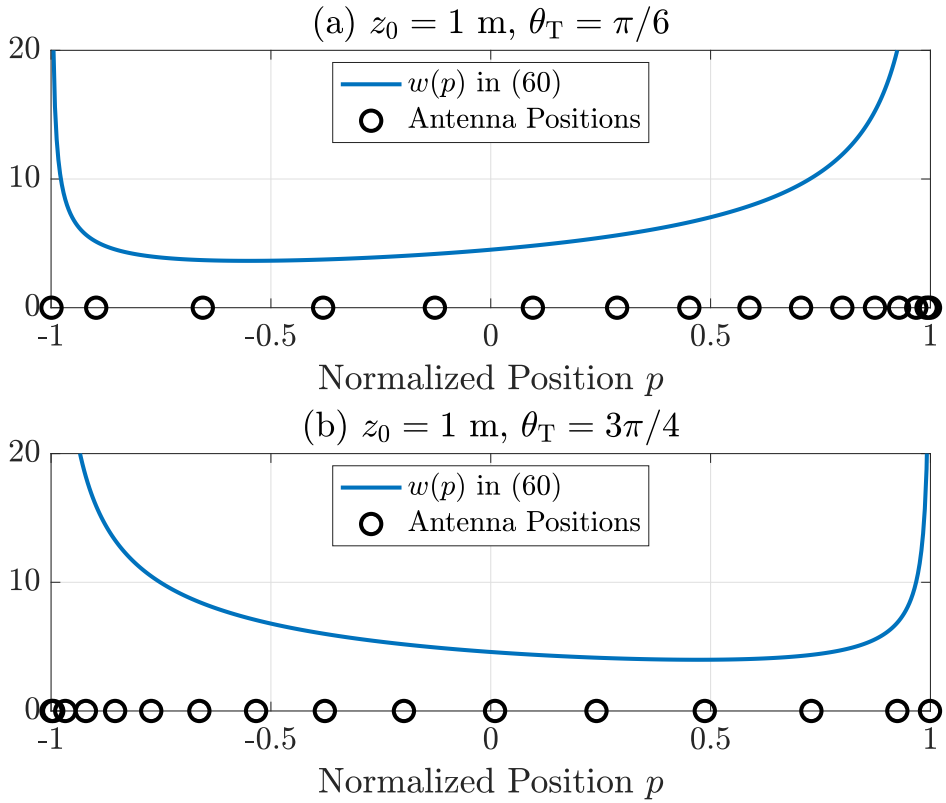
- $\alpha = 0$: ULA; $\alpha = -\frac{1}{4}$: UCA; $\alpha \to -\frac{1}{2}$: edge-heavy.

[9] M. E. Fisher and R. E. Hartwig, "Toeplitz determinants: Some applications, theorems, and conjectures," in *Advances in Chemical Physics*. John Wiley & Sons, Ltd, 1969, pp. 333–353.

[10] A. Lenard, "Some remarks on large Toeplitz determinants.," *Pacific Journal of Mathematics*, vol. 42, no. 1, pp. 137–145, 1972.

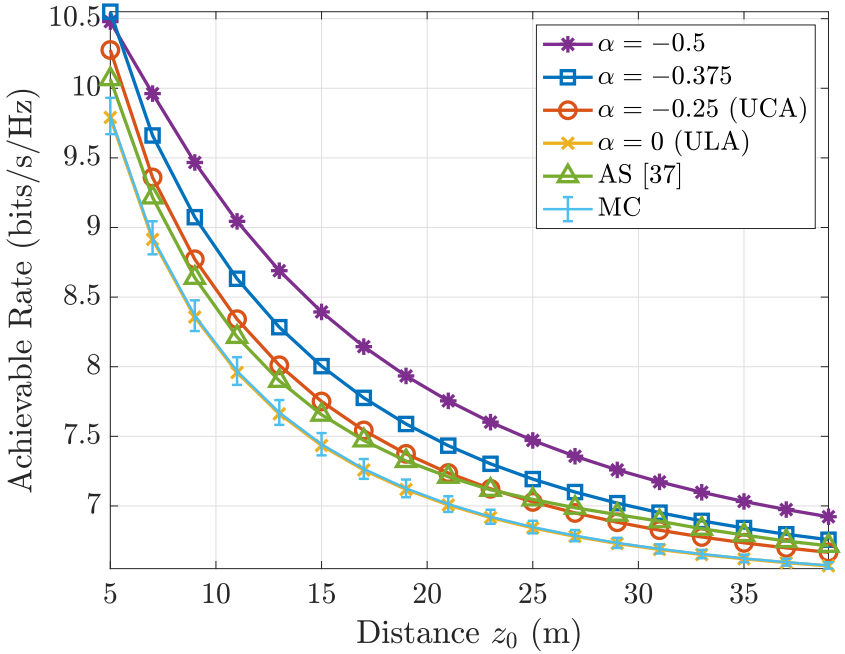
A Glimpse of Antenna Density

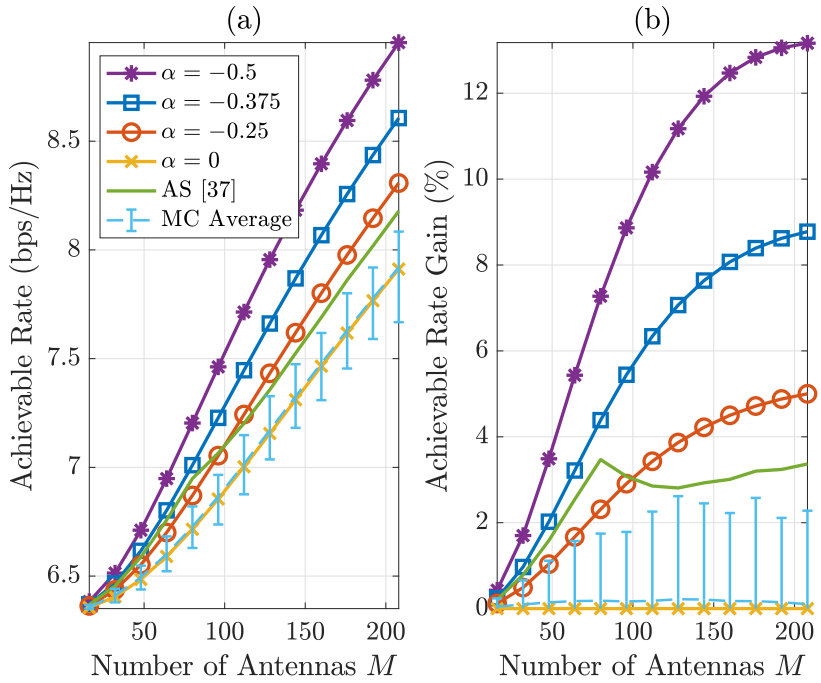
- Optimal ADF (and corresponding positions) for different incident directions
- Antennas tend to cluster towards the incident direction



Numerical Results

- Antenna positions have significant effect on achievable rate
- By simply adopting UCA, achievable rate can be improved
- Up to 13% performance gain by closed-form solution





Outline

1 Background

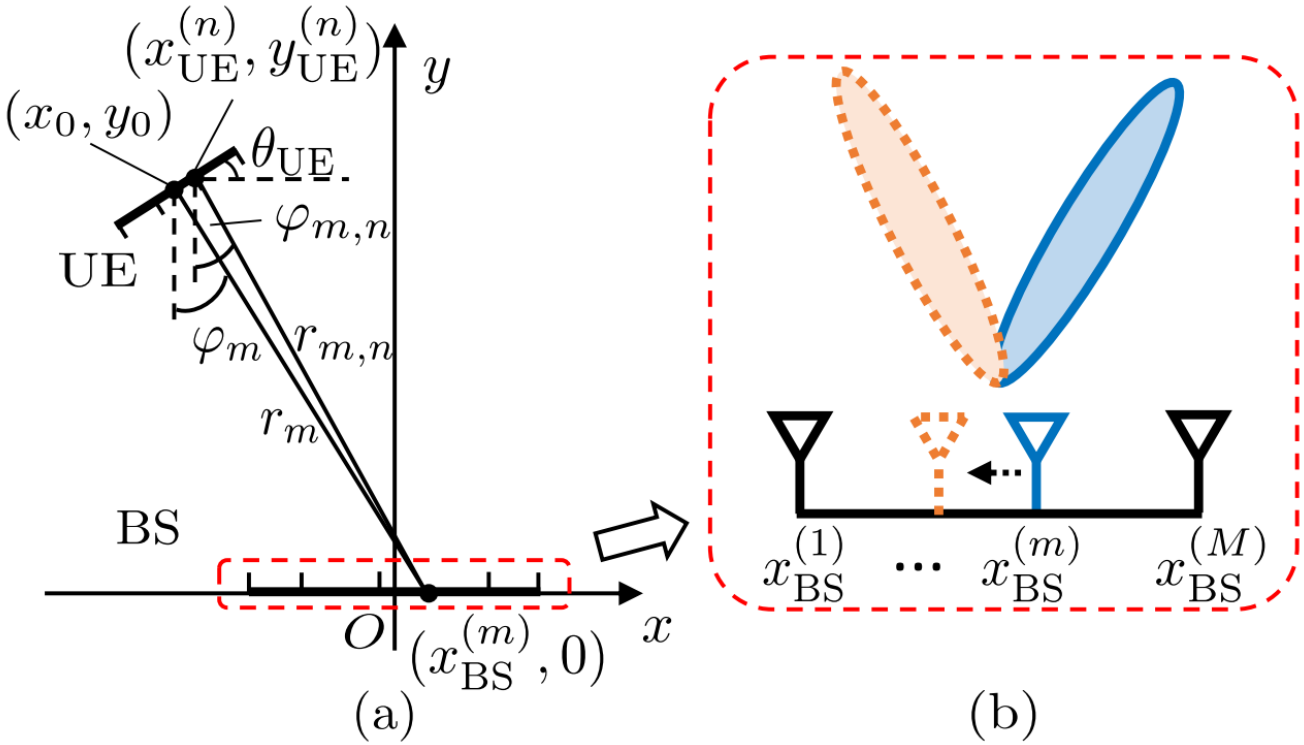
2 Antenna Density (Continuous)

3 **Antenna Placement (Discrete)**

4 Conclusions

System Model

- Cartesian coordinate system with trigonometric substitution.
- M Transmit antennas.



Problem Formulation

- Maximize spectral efficiency

$$R_{\mathbf{x}} = \log_2 \det \left(\mathbf{I}_N + \frac{\rho}{M} \mathbf{H}_{\mathbf{x}} \mathbf{H}_{\mathbf{x}}^H \right),$$

- Under typical near-field communication settings

$$R_{\mathbf{x}} \simeq \log_2 \det \left(\mathbf{D}_{\mathrm{T}} \mathbf{D}_{\mathrm{T}}^H \right) + \log_2 \det \left(\left(\underline{\mathbf{V}}_{\mathrm{T}} \right)^H \underline{\mathbf{V}}_{\mathrm{T}} \right) + \text{const},$$

where

$$\mathbf{D}_{\mathrm{T}} = \operatorname{diag} \left(\left[\frac{e^{\jmath \kappa r_1}}{r_1}, \frac{e^{\jmath \kappa r_2}}{r_2}, \cdots, \frac{e^{\jmath \kappa r_M}}{r_M} \right] \right) \quad \Leftarrow \text{Diagonal}$$

$$\mathbf{V}_{\mathrm{T}} = \begin{bmatrix} 1 & \sin\left(\varphi_1 - \theta_{\mathrm{UE}}\right) & \sin^2\left(\varphi_1 - \theta_{\mathrm{UE}}\right) & \cdots \\ 1 & \sin\left(\varphi_2 - \theta_{\mathrm{UE}}\right) & \sin^2\left(\varphi_2 - \theta_{\mathrm{UE}}\right) & \cdots \\ \vdots & \vdots & \vdots & \cdots \\ 1 & \sin\left(\varphi_M - \theta_{\mathrm{UE}}\right) & \sin^2\left(\varphi_M - \theta_{\mathrm{UE}}\right) & \cdots \end{bmatrix} \quad \Leftarrow \text{Vandermonde}$$

Problem Formulation

- Vandermonde determinant maximization problem $\Rightarrow$ ***Weighted Fekete problem*** [11], [12]

$$\begin{aligned} \max_{\mathbf{s}} \quad & 2 \sum_{1 \le i < j \le M} \log_2 (s_j - s_i) + \sum_{m=1}^{M} \log_2 \left(1 - \tilde{s}_m^2\right) \\ \text{s.t.} \quad & s_{\min} = s_1 < s_2 < \cdots < s_M = s_{\max}, \end{aligned}$$

where $s_m = \sin(\varphi_m - \theta_{\mathrm{UE}})$ and $\tilde{s}_m = \sin \varphi_m = \sin(\arcsin s_m + \theta_{\mathrm{UE}})$

- Proved to be *concave*
- Provide clear physical insights

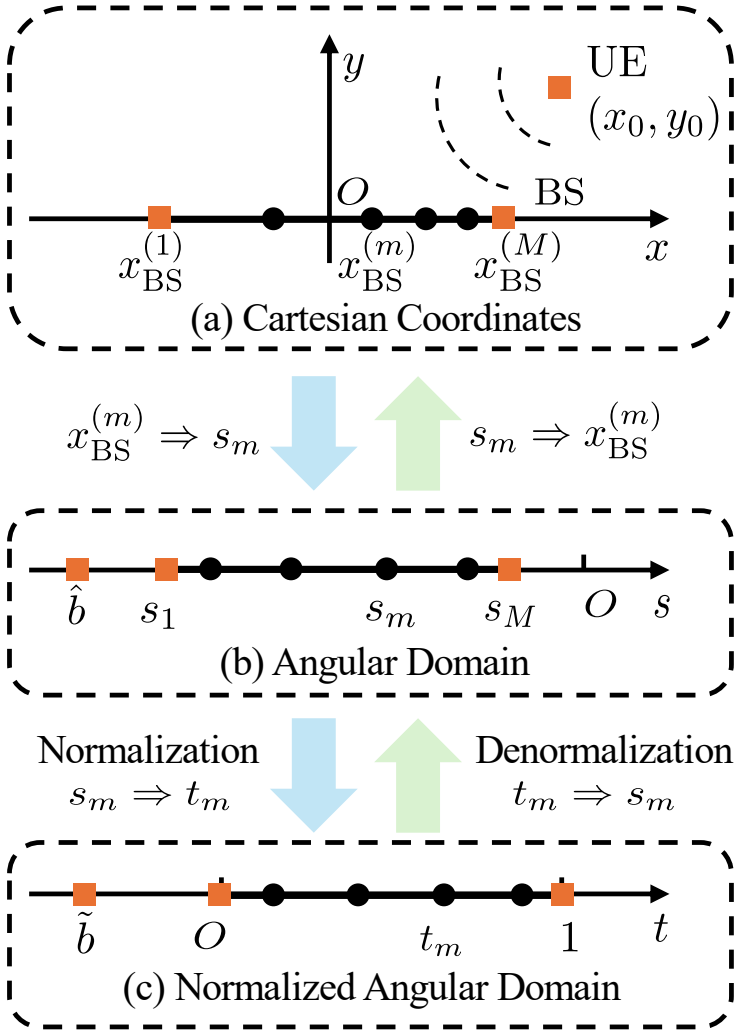
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[11] E. B. Saff and V. Totik, *Logarithmic Potentials with External Fields* (Grundlehren der mathematischen Wissenschaften), Second. Cham, Switzerland: Springer, 1997, vol. 316, ISBN: 978-3-031-65133-5.

[12] P. Wang, Y. Li, Y. Peng, S. C. Liew, and B. Vucetic, "Non-uniform linear antenna array design and optimization for millimeter-wave communications," *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7343–7356, 2016.

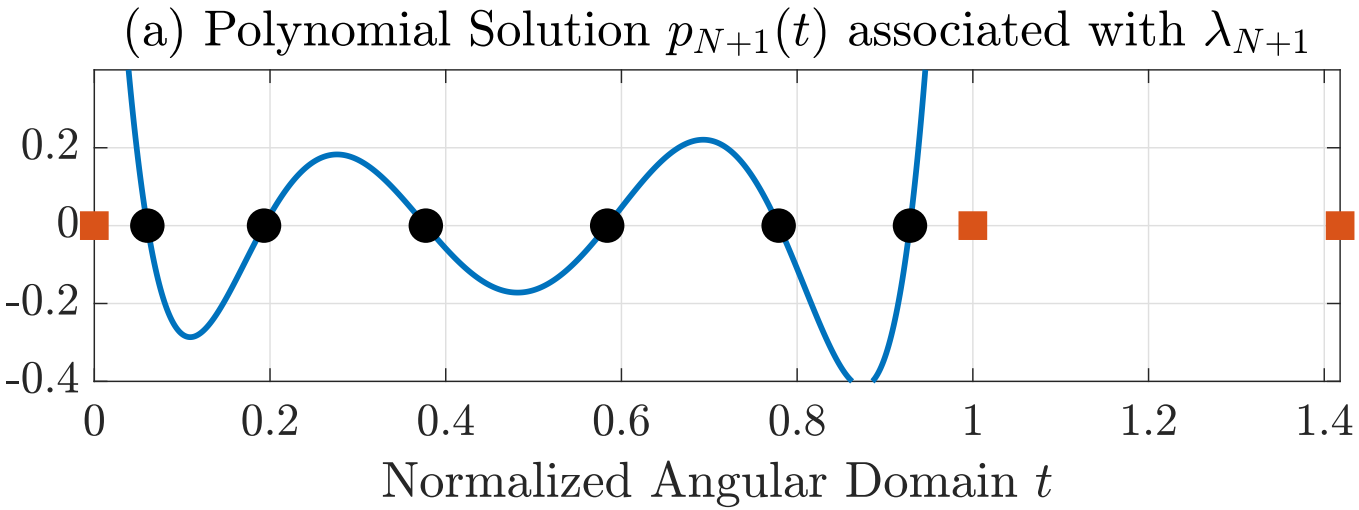
Physical Insight: Electrostatic Equilibrium

- **Free charges**: $M-2$ antennas, mutual repulsion
- **Fixed charges**: two edge antennas
- **External charge**: determined by UE position
- Optimal placement $\Leftrightarrow$ Equilibrium



Analytical Solution

- The positions of charges are the zeros of the polynomial solution to **Heine-Stieltjes ODE**.
  - In near-field LoS scenario, it degenerates to Heun equation;
  - In far-field region, it further degenerates to Jacobi equation
- $M = 8$ case:



Numerical Solution: Two-Step EVD

- Take Heun equation as example:

$$p''(t)+\left(\frac{\gamma}{t}+\frac{\delta}{t-1}+\frac{\epsilon}{t-a_{\mathrm{H}}}\right)p'(t)+\frac{\alpha\beta t-\lambda}{t(t-1)(t-a_{\mathrm{H}})}p(t)=0,$$

(1) Build matrix $\mathbf{R}$ with Heun parameters, find eigenvector $\boldsymbol{q}$

$$\mathbf{R}=\begin{bmatrix} B_0 & C_0 & 0 & \cdots & 0 \\ A_1 & B_1 & C_1 & \ddots & \vdots \\ 0 & A_2 & B_2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & C_{M-3} \\ 0 & \cdots & 0 & A_{M-2} & B_{M-2} \end{bmatrix} \in \mathbb{R}^{(M-1)\times(M-1)} \quad \Rightarrow \quad \boldsymbol{q}$$

(2) Build matrix $\mathbf{C}$ with $\boldsymbol{q}$

$$\mathbf{C}=\begin{bmatrix} \mathbf{0}_{1\times(M-3)} & -\dfrac{\boldsymbol{q}_{0:M-3}}{\boldsymbol{q}_{M-2}} \\ \mathbf{I}_{M-3} & \end{bmatrix} \in \mathbb{R}^{(M-2)\times(M-2)} \quad \Rightarrow \quad \boxed{\text{Zeros of } p(t)}$$

- See Appendix and [8] for more details.

[8] S. Liu, X. Yu, S. Song, and K. B. Letaief, "Near-field communication with movable antennas: An electrostatic equilibrium perspective," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 16 024–16 039, 2026.

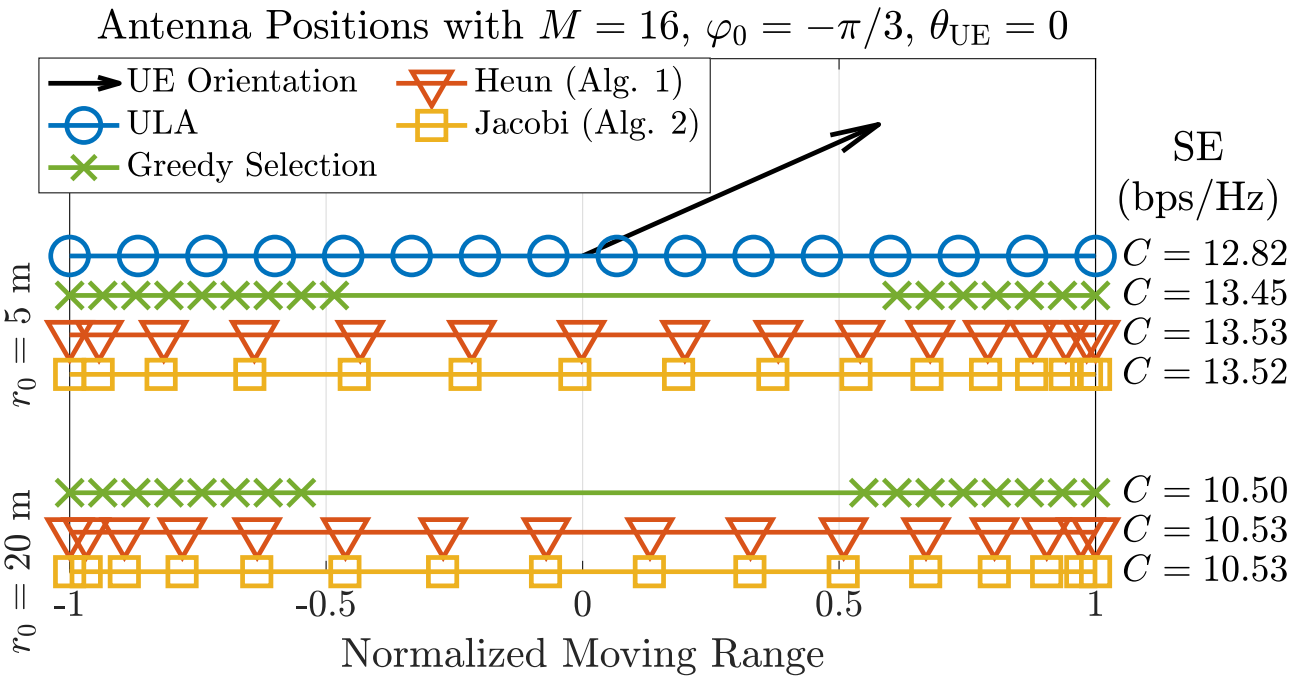
Numerical Solution: Closed-Form Approximation

- When Tx and Rx are not extremely close to each other, the effect of external field can be ignored.
- Degenerate to Jacobi case. Admit close-form solution

$$t = \cos\left(\frac{(2n+1)\,\pi}{2N}\right), \quad n = 0, 1, \cdots, N-1;\ N = M-2.$$

Numerical Results

- Denser at both edges, shifted toward the UE side (aligns with continuous analysis).
- $r_0 = 5\,\text{m}$: Heun $\approx$ Jacobi $>$ Greedy $>$ ULA ($+5.5\%$ vs. ULA)
- $r_0 = 20\,\text{m}$: Jacobi (closed form) $=$ Heun (EVD)



Outline

1 Background

2 Antenna Density (Continuous)

3 Antenna Placement (Discrete)

4 **Conclusions**

Conclusions

Take-home Conclusions:

- **Large** M: Antennas should cluster towards edges, i.e., U-shaped density.
- **Moderate** M: Precise positions are the equilibrium points in the angular domain.

[1] L. Zhu, W. Ma, and R. Zhang, "Movable antennas for wireless communication: Opportunities and challenges," *IEEE Wireless Commun.*, vol. 62, no. 6, pp. 114–120, 2024.

[2] K.-K. Wong, A. Shojaeifard, K.-F. Tong, and Y. Zhang, "Fluid antenna systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1950–1962, 2021.

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[6] X. Xu, X. Mu, Y. Liu, and A. Nallanathan, "Joint transmit and pinching beamforming for pinching antenna system (pass): Optimization-based or learning-based?" *IEEE Trans. Wireless Commun.*, vol. 25, pp. 11 449–11 464, 2026.

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[10] A. Lenard, "Some remarks on large Toeplitz determinants.," *Pacific Journal of Mathematics*, vol. 42, no. 1, pp. 137–145, 1972.

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[12] P. Wang, Y. Li, Y. Peng, S. C. Liew, and B. Vucetic, "Non-uniform linear antenna array design and optimization for millimeter-wave communications," *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7343–7356, 2016.

[13] R. Berman, S. Boucksom, and D. W. Nyström, "Fekete points and convergence towards equilibrium measures on complex manifolds," *Acta Mathematica*, vol. 207, no. 1, pp. 1–27, Sep. 2011, ISSN: 1871-2509.

Thanks for your attention!

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Codes & Slides: https://scliubit.github.io

Toeplitz Matrix

Definition 1: Generating Function of Toeplitz Matrix

Let $s(\theta)$ be a real smooth and non-vanishing function defined on unit circle with Fourier coefficients

$$c_k = \frac{1}{2\pi}\int_0^{2\pi} s(\theta)e^{-\jmath k\theta}\,\mathrm{d}\theta, \quad k \in \mathbb{Z}.$$

If a Toeplitz matrix $\mathbf{T}_N$ admits the form as

$$\mathbf{T}_N = \begin{bmatrix} c_0 & c_{-1} & c_{-2} & \cdots & c_{1-N} \\ c_1 & c_0 & c_{-1} & \cdots & c_{2-N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_{N-1} & c_{N-2} & c_{N-3} & \cdots & c_0 \end{bmatrix},$$

then $s(\theta)$ is called the **generating function** of $\mathbf{T}_N$.

Heun

$$\begin{cases} A_n = (n-1+\alpha)(n-1+\beta) \\ B_n = -n\left((n-1+\gamma)(1+a_{\mathrm{H}}) + a_{\mathrm{H}}\delta + \epsilon\right) , \\ C_n = (n+1)(n+\gamma)a_{\mathrm{H}} \end{cases}$$

Multi-User Case

- For multi-user case, we resort to conventional Cartesian coordinates. The objective function can still be formulated into an equilibrium form, while the approximation error may become slightly higher.
- For the k-th user, target and distance

$$R_{\mathbf{x}}^{(k)} = \log_2 \det \left(\mathbf{I} + \rho \mathbf{H}_{\mathbf{x}}^{(k)} \left(\mathbf{H}_{\mathbf{x}}^{(k)} \right)^H \right),$$

$$\begin{aligned} r_{m,n,k} &\simeq \sqrt{r_{m,k}^2 + \left(x_{\mathrm{BS}}^{(m)} \right)^2 + 2 d_n^{(k)} \left(x_{\mathrm{BS}}^{(m)} - x_0^{(k)} \right) \cos \theta_{\mathrm{UE}}^{(k)}} \\ &\simeq r_{m,k} + \frac{d_n^{(k)} \cos \theta_{\mathrm{UE}}^{(k)} \left(x_{\mathrm{BS}}^{(m)} - x_0^{(k)} \right)}{r_0^{(k)}}, \end{aligned}$$

Multi-User Case

- User-specific Vandermonde matrix (k omitted for brevity)

$$\mathbf{V}_{\mathrm{T}} = \begin{bmatrix} 1 & \frac{x_{\mathrm{BS}}^{(1)} - x_0}{r_0} & \left(\frac{x_{\mathrm{BS}}^{(1)} - x_0}{r_0}\right)^2 & \cdots \\ 1 & \frac{x_{\mathrm{BS}}^{(2)} - x_0}{r_0} & \left(\frac{x_{\mathrm{BS}}^{(2)} - x_0}{r_0}\right)^2 & \cdots \\ \vdots & \vdots & \vdots & \cdots \\ 1 & \frac{x_{\mathrm{BS}}^{(M)} - x_0}{r_0} & \left(\frac{x_{\mathrm{BS}}^{(M)} - x_0}{r_0}\right)^2 & \cdots \end{bmatrix},$$

- Overall objective the overall objective for the k-th UE can then be given by

$$\begin{aligned} J_k(\mathbf{x}) &= \sum_{1 \le i < j \le M} \log_2 \left(\frac{x_{\mathrm{BS}}^{(j)} - x_0^{(k)}}{r_0^{(k)}} - \frac{x_{\mathrm{BS}}^{(i)} - x_0^{(k)}}{r_0^{(k)}} \right) + \sum_{m=1}^{M} \log_2 \left(\frac{1}{r_{m,k}} \right) \\ &= \sum_{1 \le i < j \le M} \log_2 \left(x_{\mathrm{BS}}^{(j)} - x_{\mathrm{BS}}^{(i)} \right) + \sum_{m=1}^{M} \log_2 \left(\frac{1}{r_{m,k}} \right) + C_0^{(k)}, \end{aligned}$$

Planar Array

- The original problem will be extended to a multivariate Vandermonde determinant maximization problem.

$$\mathbf{\Sigma} = \operatorname{diag}\left(\sum_{i+j=0} \frac{(-\jmath\kappa)^0}{i!j!}, \sum_{i+j=1} \frac{(-\jmath\kappa)^1}{i!j!}, \cdots \right),$$

$$\mathbf{V}_{\mathrm{T}} = \begin{bmatrix} 1 & \alpha_1 & \beta_1 & \alpha_1^2 & \alpha_1\beta_1 & \beta_1^2 & \cdots \\ 1 & \alpha_2 & \beta_2 & \alpha_2^2 & \alpha_2\beta_2 & \beta_2^2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \cdots \\ 1 & \alpha_M & \beta_M & \alpha_M^2 & \alpha_M\beta_M & \beta_M^2 & \cdots \end{bmatrix},$$

- The physical picture goes beyond the original electrostatic equilibrium model. It is the equilibrium on complex manifolds [13]

[13] R. Berman, S. Boucksom, and D. W. Nyström, "Fekete points and convergence towards equilibrium measures on complex manifolds," *Acta Mathematica*, vol. 207, no. 1, pp. 1–27, Sep. 2011, ISSN: 1871-2509.